Lab 1 - Version 1

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Version 1

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1 Introduction

The beverage manufacturing process requires coordination between brands, suppliers, co-packagers, distributors, logistics providers, and internal operational personnel. Production information is distributed across spreadsheets, emails, phone calls, and different software systems. As the number of products and customers increase, separated information creates communication delays, approval bottlenecks, duplicated work and overall difficulty determining the current status of production. As the industry scales the importance of efficient operation coordination increases.

1.1 Purpose

Beverage manufacturing workflows depend on information being able to move accurately between multiple organizations and operations. Existing processes may separate product requests, supplier information, approvals, production schedules, shipping information, documents, and overall communications into different systems. This separation increases the efforts required to locate and process up to date information. BevOS addresses this problem by providing a centralized workflow management platform that connects related operation information within one common system.

1.2 Scope

BevOS is intended to centralize information associated with beverage manufacturing workflows rather than replace every system used. The platform supports product requests, supplier quotes, approvals, production tracking, logistics coordination, document storage, and communication. Authorized users can monitor current workflow information through a shared platform while maintaining responsibilities associated with their individual roles. The prototype focuses on demonstrating how these capabilities can reduce separated communication and improve visibility.

2 General Description

BevOS provides a shared environment for managing information throughout beverage manufacturing operations. Its main goal is to reduce dependences on disconnected applications and make operational information easier to locate and manage.

The major goals of BevOS are to improve operational efficiency, reduce communication and approval delays, increase visibility throughout the production lifecycle, organize documents and records, and support management decisions through reports and dashboards.

BevOS is intended for several types of users involved in beverage production. Brands can create product requests and monitor progress, while distributors can coordinate product availability and movement. Suppliers can receive request and provide quotes or other service information. Production. Logistics, finance, and other authorized operational personnel can contribute information needed throughout the workflow.

The major objectives and functional areas of BevOS are summarized in Figure 1.

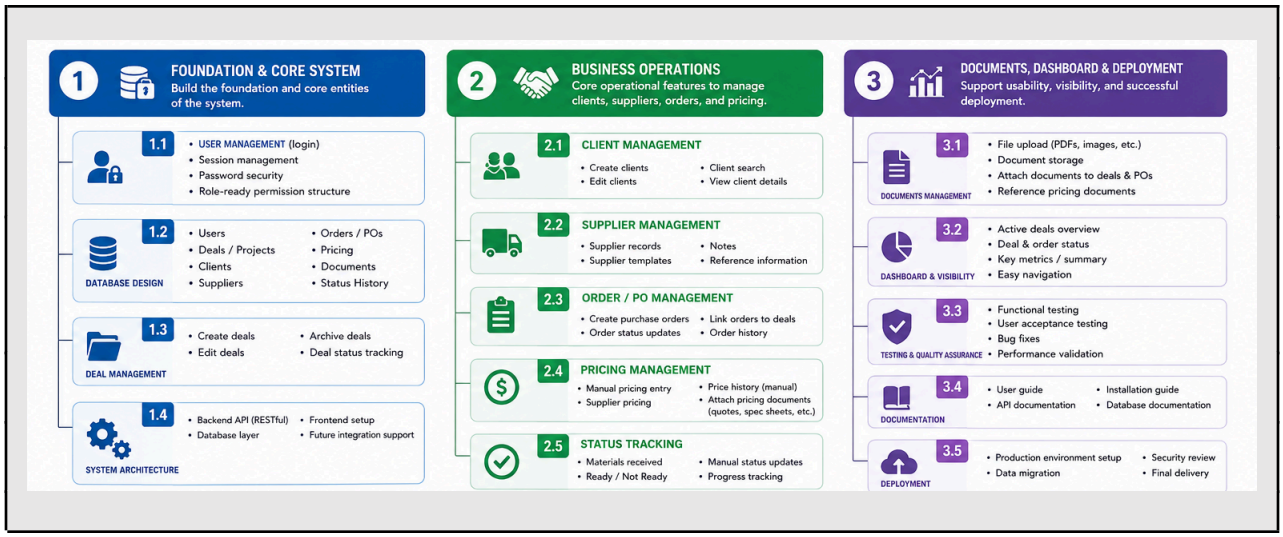


Figure 1. BevOS Objective Breakdown

2.1 Workflow and Request Management

BevOS centralizes operational activities so that requests, approvals, documentation, communication, production information, and logistics information remains connected throughout the workflow. This structure reduces the chance that information becomes separated across different spreadsheets, email conversations, or unrelated applications.

Product Request Management allows authorized users to create, review and monitor product requests within a consisted process. Request details and status information remain available through workflow, reducing the risk of incomplete or lost requests.

Quote Management supports supplier quote requests and keeps supplier responses associated with the appropriate product request. Keeping this information together improves visibility during supplier selection while reducing repeated follow ups.

Approval Tracking manages artwork, purchase orders, and other workflow approvals. Approval status and history can be recorded so that there is a record of past and active tasks. This allows users to identify approval bottlenecks before causing unnecessary delay.

2.2 Production, Logistics, and Document Management

Production Management connects approved work with manufacturing activities by tracking schedules, goals, progress updates, and status. Keeping this information up to date helps operational teams identify changes or delays before they occur.

Logistics Coordination continues the workflow after production by connecting production completion with transportation and delivery information. Shipping activity and delivery status can be tracked so that authorized stakeholders can determine where a product is at in the process.

Document Management stores operational files and stores them with appropriate requests or workflow. These files may include artwork, purchase orders, production documents, or shipping records. Centralized storage reduces misplacement or lost documents. Access to these documents can also be limited according to user responsibility.

2.3 Communication, Reporting, and Issue Management

The Communication Platform keeps workflow related messages and updates connected to the activities they pertain to. Communication may be associated with a request, quote, approval, production, or shipment.

Reporting and Dashboard features summarize operational status, workflow activity, delays, and other performance information. Different users can receive views appropriate to their responsibilities, allowing stored operational data to support monitoring and decision making. Some key components to the dashboard are shown below in figure 2.

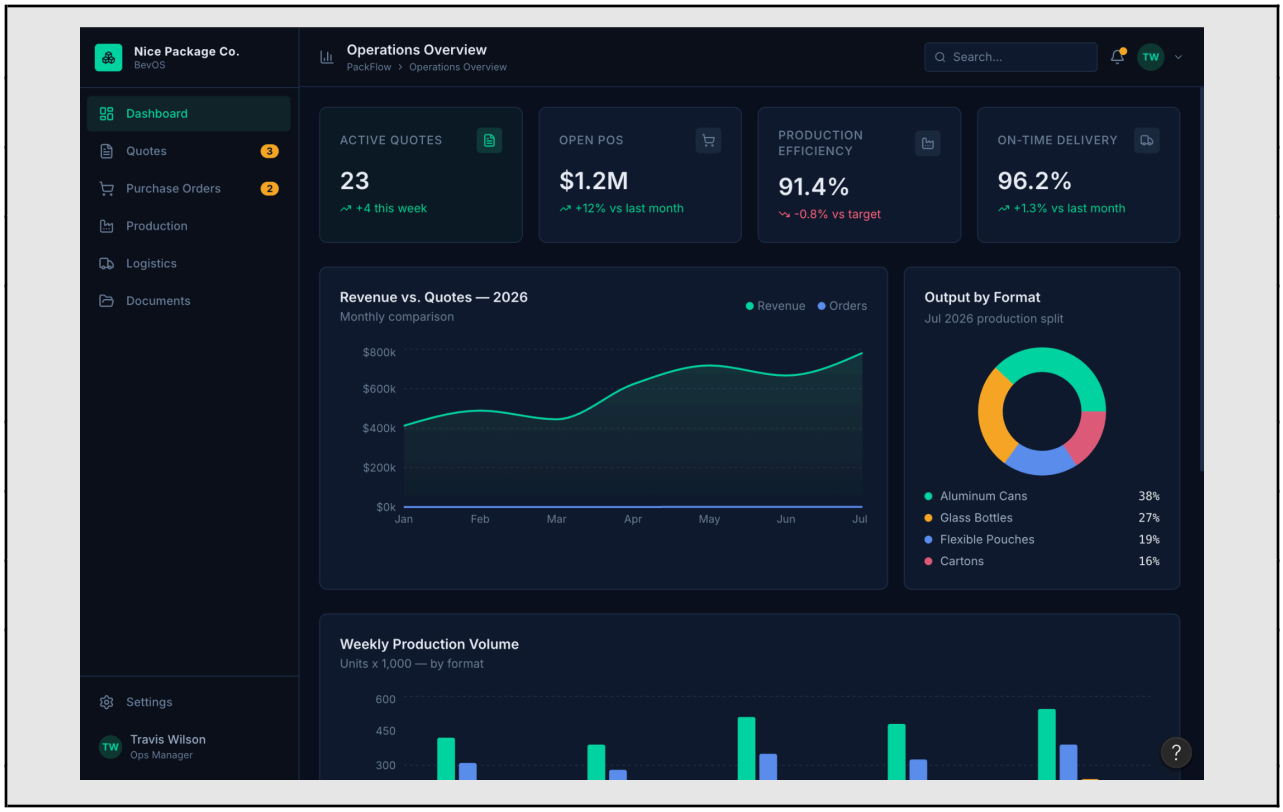


Figure 2. BevOS Dashboard

Issue Tracking records workflow problems, and resolution details. Recurring problems and user feedback can then be reviewed to identify weaknesses within a process.

2.4 Access Control and Accountability

BevOS limits information and system functions according to each users role and their responsibilities. Role based access helps protect confidential information while allowing authorized personnel to complete the required tasks.

The platform also maintains records of important workflow changes and decisions. These records improve traceability and provide greater accountability when users need to determine when information changed or which actions were completed.

2.5 Major Components

BevOS uses a web based architecture that separates the user interface, shared system serves, specialized operational modules, and stored data. The prototype does not require specialized manufacturing hardware because BevOS is managing workflow rather than directly controlling production equipment.

2.5.1 Hardware Components

Users access BevOS from desktop or laptop computers that are capable of running modern web browsers. Server resources support the application software, database, and document storage functions. Because BevOS focuses on workflow and information management, specialized production machinery is not required to operate the prototype.

2.5.2 Software Components

The BevOS software architecture includes a web user interface, Core Serves operational modules, document storage, reporting functions, and a PostgreSQL database.

The web user interface provides the primary point of interaction for brands, distributors, suppliers, and other authorized users. Requests entered through the interface are sent to BevOS core services, which applies workflow rules, manages access requirements, and coordinates the system's functional modules.

Specialized modules handle quotes, communications, production, logistics, document storage, and reporting. PostgreSQL stores structured application information such as users, requests, quotes, approvals, production records, logistics information, and reporting data.

2.5.3 Data Flow

Users begin an interaction with BevOS through the web interface. The interface sends the request to BevOS Core Services, where system rules and access requirements are applied. Core Services then coordinates the appropriate operation module, such as quotes, production, logistics, communications, reporting, or documentation storage.

The functional modules use PostgreSQL to store and retrieve the structured information needed to complete their responsibilities. Results are returned through the application services and displayed to the user through the BevOS interface.

Separating user interaction, shared application logic, operational functions, and consistent data storage make the system easier to maintain and expand. Individual modules can change as business needs while continuing to use the same core services and database structure.

3 Glossary

- **Approval Workflow** - A process used to review and approve operational tasks before they move to the next stage.
- **BevOS** - A centralized workflow-management platform designed for beverage manufacturing operations.
- **Client** - An organization or individual purchasing services or products from another participant in the manufacturing process.
- **Co-Packer** - A third-party company that manufactures and packages beverages for another company.
- **Core Services** - The shared BevOS system software layer that processes requests, applies system rules and coordination functional modules, and manages data exchange.
- **Dashboard** - A visual interface displaying operational information and performance metrics.
- **Distributor** - An organization responsible for delivering products from manufacturers to retailers or customers.
- **ERP (Enterprise Resource Planning)** - Software that integrates core business processes into a unified system.
- **Logistics** - The planning and coordination of transportation and delivery activities.
- **MFCD (Major Functional Components Diagram)** - A diagram illustrating the primary hardware and software components of the system.
- **Nice Package Company** - The beverage-industry client and domain partner for whom BevOS is being developed.
- **PostgreSQL** - An open-source relational database management system used to store application data.
- **Role-Based Access Control** - A security method that grants system permissions according to the user's assigned role and responsibilities.
- **Single Source of Truth** - A trusted, centralized set of current information used consistently by authorized stakeholders.
- **Stakeholder** - A person or organization that participates in or is affected by a beverage-manufacturing workflow.
- **Workflow** - A sequence of business activities completed to achieve a specific objective.

4 References

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